

Retrieval-augmented generation often suffers from hallucinations and misattributed citations where ge

Verbatim text span alignment maps retrieved chunk elements directly to PDF selectable text layer cha

Hybrid retrieval combines dense vector embeddings with PostgreSQL tsvector full-text search, merged

Multilingual cross-encoder reranking re-evaluates top 40 candidate chunks to distill the top 6 evidence

Bounding box origin normalization converts PDF bottom-left points into standard top-left DOM coordin

Claim-to-citation validation checks semantic support between the generated assertion and the reference